

Pramo Jewels

Performance Testing Plan

Document ID: PJ-PTP-001

Version: 1.0

Status: Approved for Development

Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Define the performance testing approach, workloads, success criteria and monitoring
Audience	QA Engineers, Performance Testers, Developers, DevOps, Product Owners
Version	1.0

Performance Testing Scope

Test Type	Objective
Load Testing	Validate expected concurrent user capacity
Stress Testing	Identify breaking points under extreme load
Spike Testing	Measure recovery after sudden traffic spikes
Endurance Testing	Detect memory leaks during prolonged execution
Scalability Testing	Verify horizontal and vertical scaling
API Performance	Measure latency, throughput and error rates
Database Performance	Validate query execution and indexing
Frontend Performance	Assess page load and Core Web Vitals

Objectives

Ensure the platform remains responsive, reliable and scalable during normal and peak business traffic.

Workloads

Simulate browsing, product search, checkout, payment, order tracking, admin operations and API traffic.

Environment

Execute tests in a staging environment closely matching production infrastructure and configurations.

Monitoring

Collect CPU, memory, disk, network, database, cache, API latency, throughput and error-rate metrics.

Success Criteria

Meet defined response-time SLAs, maintain acceptable error rates, support target concurrency and recover gracefully after spikes.

Reporting

Provide execution summaries, bottleneck analysis, trend comparisons, recommendations and optimization opportunities.

Risk Management

Focus on checkout, payments, authentication, inventory synchronization and third-party service dependencies.

Acceptance Criteria

Performance objectives are achieved, no critical bottlenecks remain unresolved and stakeholders approve release readiness.